Course Overview

COSC3060 - Web Programming Studio

School of Science, Engineering & Technology
RMIT University Vietnam

Today agenda

1. Course syllabus
2. Team formation
3. Assignment

Course syllabus

Week	Module	Assessments
1	Week 1. HTML 1: Structure & Content; Innovation: Project Ideation	
2	Week 2: HTML 2: User Input & CSS 1: Syntax & Formatting	
3	Week 3. CSS2: Page Layout; Wire framing; Javascript: Fundamentals	
4	Week 4. Javascript: DOM, String processing	Due-date for Assessment Task 1
5	Week 5. Javascript: Forms & APIs	
6	Week 6. Introduction to NodeJS	
7	Week 7. Independent learning week	
8	Week 8. NoSQL Database, MongoDB, Entity Relationships in NoSQL Database	Due-date for Assessment Task 2
9	Week 9. Usability Testing; Web Data Storage	
10	Week 10. Storage of External Files; NodeJS with HTML APIs	
11	Week 11 - Mock Exam, Product Demonstration	
12	Week 12: Review and Final Exam	Due-date for Assessment Task 4

Course Learning Components

Full-stack Development	Front-end Development	Web pages	HTML + CSS + Javascript	W1, 2, 3, 4
	Back-end Development	Server	NodeJS + Express JS	W5, 6, 8
		Database	MongoDB	W9,10,11

Assessments & Timeline

Assessment 1 - Studio Project- Design Document & Static Front-End Web pages — 20% of Total

- **Assessment Task 1**
  Due Jul 24 at 23:59 | 20 pts

Assessment 2 - Studio Project – Prototype — 20% of Total

- **Assessment Task 2**
  Due Aug 21 at 23:59 | 20 pts

Assessment 3 - Studio Project – Final Product Demonstration and Report — 20% of Total

- **Assessment Task 3.1: Complete Product**
  Due Sep 11 at 23:59 | 10 pts
- **Assessment Task 3.2: Product Demonstration and Report**
  Due Sep 11 at 23:59 | 10 pts

Assessment 4 - Fullstack in-class Lab Test — 40% of Total

- **Assessment Task 4**
  Week 12: Review and Final Exam Module | **Available until** Sep 18 at 13:30 | **Due** Sep 18 at 11:00 | 40 pts

Learning outcomes

CLO1: Write web programming code and access data through the adoption of accepted standards, mark-up languages, client-side programming, and server-side programming

CLO2: Design and implement interactive web sites for medium-sized applications with regard to issues of usability, accessibility and internationalisation

CLO3: Design and implement a medium-sized client-server internet application that accommodates specific requirements and constraints, based on analysis, modelling or requirements specification

CLO4: Demonstrate skills for self-directed learning, reflection and evaluation of your own and your peers work to improve professional practice

CLO5: Demonstrate adherence to appropriate standards and practice of Professionalism and Ethics in designing and implementing web solutions.

Learning in studio mode

Studio Mode is learning by doing, creating, and collaborating just like in a real web development team.

Two 3-hour sessions per week:

- Concept & Practice — learn and experiment with new tools or topics.
- Studio Session — apply knowledge directly to your group project.
- Expect hands-on teamwork, feedback, and iteration each week.

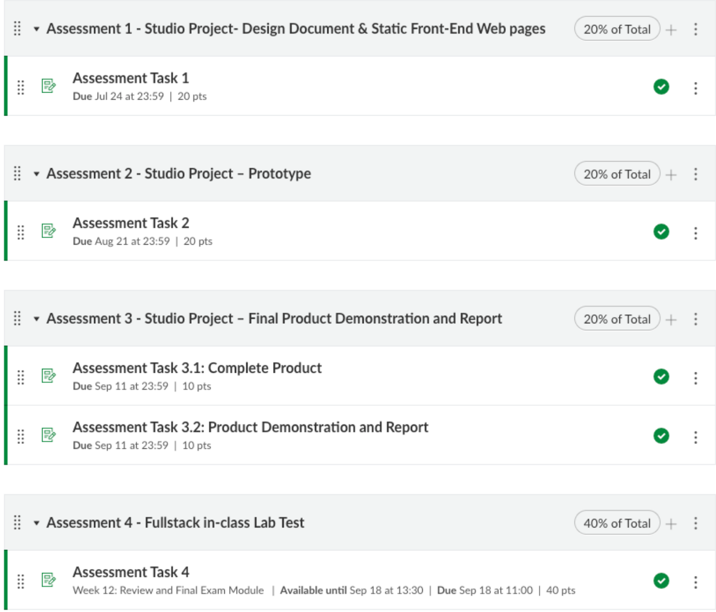
Before each session, you should:

- Review the weekly materials and announcements on Canvas.
- Complete preparation activities or coding practice at home.
- Come to class, prepared with questions, ideas, and partial work to discuss.

Team Formation

- Group project is a must for this course.
- All teams must be formed and confirmed in week 1.

  Otherwise, we will not have efficient time for the project completion.
- Each team has 4 members of the same class groups.
- Due to the current number of students, a few exceptions are:
  - ✓ SGS-Group 1 and 2 have one team of 3 members
  - ✓ HN has one team of 5 members
- How to sign up to a team:
  - ✓ By self sign-up on Canvas



Self sign-up to a team

Step 1. Make friends

- Get to know friends in class, have a small talks, share your expertise and project.

Step 2. Form a team of 4 members and self sign-up on Canvas

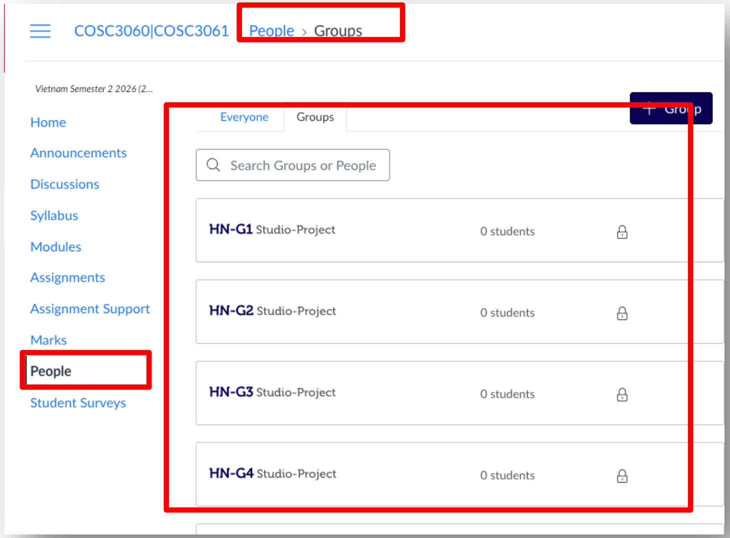
- Access Canvas shell → People → All members of your team choose a team and self sign-up to it.
- Choose the team in your class group
  - ✓ SGS1- is for class group 1 at SGS Campus
  - ✓ SGS2- is for class group 2 at SGS Campus
  - ✓ HN- is for HN campus

Step 3. Fill team members' information in the group file

- Access Canvas shell → Modules → Group Registration → Group Information

Note.

- Only one 3-member team is allowed for SGS1- and SGS2-
- Only one 5-member team is allowed for HN-



Communication

In person: during classes (prior appointment required)

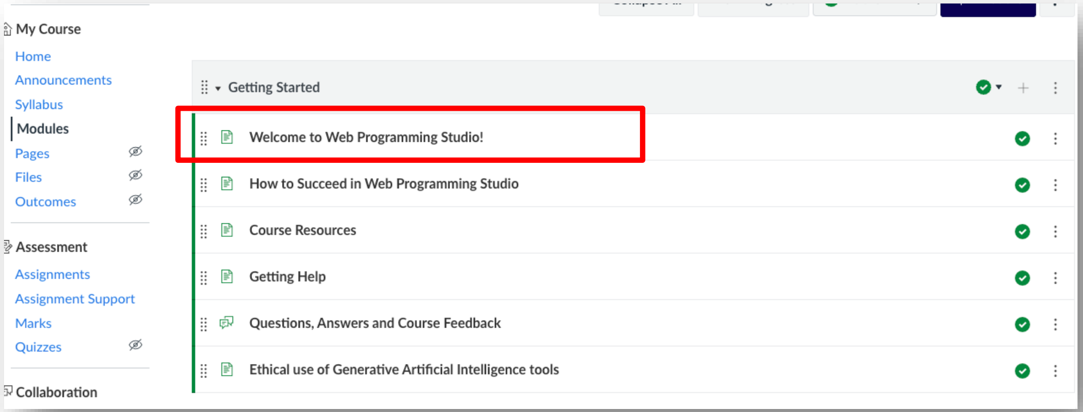
Discussions forum on Canvas: to ask questions about course materials, lab exercises, etc.

Email: to discuss with me any personal issues which requires my attention.

The title of the email need contain **“[COSC3060]” or “[COSC3061]”.**

Note: please **do not** ask questions via Ms. Teams.

Contact of Teaching staff:



Group assignment

Group assignment

Goal

To collaboratively design and develop a full-stack web application that demonstrates both technical proficiency and teamwork skills

Objectives

1. Apply HTML, CSS, JavaScript, Node.js, Express, and MongoDB to build a functional web application.
2. Practice team-based development using GitHub Classroom.
3. Design and integrate modular features
4. Demonstrate UI/UX design consistency across modules.
5. Develop project planning and collaboration skills through a shared charter and workflow.
6. Engage in peer feedback and reflective learning through studio activities.

Project Structure Overview

Five Core Modules

Each team will build an integrated web application with these components:

- 🛒 Shopping Cart
- 💬 Discussion Forum
- ✍ Blog
- ⭐ Product Review & Rating
- ❤️ Wishlist
- 🔗 Shared Module *(jointly developed by all members)*

Project phases

Three phrases

Phase 1. Static web pages(Front-End)

Design and implement all pages using HTML, CSS, and sample content.

Phase 2. Dynamic Prototype

Add interactivity and functionality using JavaScript, Node.js

Phase 3. Fully functional Application, Presentation, Report

Integrate modules, ensure consistent UI/UX, conduct testing, and prepare for demo and report submission.

Group assignment

- ✓ Comprises of three assessments: Task 1, 2, and 3
- ✓ This assignment includes both group work and individual work.
- ✓ You will work in teams of four member.
- ✓ Your performance will be assessed on:
  1. How effectively your team collaborates to produce a unified product
  2. The quality of your individual technical contribution.

Group assignment

Group Work

Your team is responsible for:

- Creating a Group Charter (shared expectations & working rules).
- Developing a Production & Development Plan (project roadmap).
- Ensuring consistent design & functionality across all modules.
- Building a shared module, with equal contributions from everyone.

These teamwork elements are part of your assessment.

Group assignment

Each member will **lead one module**:

- 🛒 Shopping Cart
- 💬 Discussion Forum
- ✍️ Blog
- ⭐ Product Review & Rating
- ❤️ Wishlist

You will be assessed individually on:

1. Technical implementation (planning, coding, functionality, documentation)
2. Professional practice (communication, collaboration)
3. Peer feedback and contribution in studio sessions

Full stack In-class Lab Test

Full-stack web application

- Within 2 hours to 2.30 hours, you will build a fully functional full stack web application in according to requirements provided.
- You are allowed to refer to course materials and your notes and will use Lab Computers during the test. You are not allowed to use AI tools or internet access during the test.

Full-stack Development

- **Front-end Development**
  - Web pages — HTML + CSS + Javascript
- **Back-end Development**
  - Server — NodeJS + Express JS
  - Database — MongoDB